

Hybrid Deep Learning Perspective on Stock Market Liquidity Prediction in the MENA Region

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Abstract—The Middle East and North Africa (MENA) region is a fascinating geographical expanse that offers a multitude of unexplored possibilities and unique obstacles. This region is becoming increasingly significant, not just because of its plentiful reserves but also because of its developing and vibrant financial market, which offers a promising basis for financial innovation and economic expansion. In the intricate world of finance, the concept of liquidity plays a vital role in ensuring the seamless functioning of markets and instilling trust in investors, and traditional models frequently fall short of accurately forecasting market liquidity, leading to a lack of comprehensive financial research and subpar investment strategies in this domain, which can be attributed to various factors, including volatility, lower liquidity compared to global standards, and the presence of distinct socio-economic elements. Our paper aims to analyze historical time series data from the MENA region and investigate the potential of a hybrid deep learning model to enhance the accuracy of MENA stock market liquidity predictions. This research can have significant implications for the development of more robust trading strategies by accurately forecasting changes in liquidity and uncovering the intricacies of market behavior.

Index Terms—MENA, stock market, liquidity, trading, investment, prediction, Hybrid deep learning,

I. INTRODUCTION

The vitality of the security market is closely tied to liquidity. If a security market has better liquidity, it can attract more investors, increase investor confidence, withstand external shocks, reduce systemic risk, and ensure stability in the security market [1] which is more challenging for the MENA structure, primarily stemming from its complicated tapestry of economic policies, volatile oil prices, and the complex structure of geopolitical influences, which have impeded the efficacy of conventional analysis methods in accurately predicting liquidity outcomes, thereby hindering the enhancement of market foresight, this gap has prompted a transition towards more sophisticated and detailed analytical instruments, at this stage, where the potential of deep learning assumes prominence.

Nevertheless, the investigation of deep learning models in this particular region is not without its difficulties.

The challenges faced in this field involve navigating through regulatory frameworks, understanding market dynamics, overcoming limited data accessibility, and addressing a lack of transparency in the MENA region. These factors contribute to

the objective's complexity and necessitate specific strategies to connect with existing methods. It is crucial to offer valuable insights and methodologies that are tailored to the distinctive market characteristics of the region. Engaging in this practice contributes to an expanding body of literature that recognizes the potential of sophisticated analytical methodologies in the field of financial stock market prediction.

II. RELATED WORKS

Formerly, statistical analytics approaches were considered the standard for time series data prediction. In contemporary times, as the complexity and amount of data have increased, artificial intelligence has emerged as the dominant force.

Artificial intelligence (AI) and its different branches, specifically deep learning (DL), have become more widely recognized due to their effective resolution of prevailing challenges in liquidity prediction by combining their exceptional capacity to analyze massive datasets and identify complex patterns and correlations that would pose significant challenges using conventional analytical methods. According to the existing literature, deep learning models outperformed conventional analytical models in the market's liquidity prediction, which is proven in the studies conducted by [6]. Four deep learning architectures, namely MLP, RNN, LSTM, and CNN, were used to forecast stock markets. Out of these designs, CNN showed superior accuracy compared to the other three networks. This is due to CNN's ability to efficiently capture unexpected changes in the system by using a precise window to forecast the next moment.

However, they missed investigating the possible benefits of using a hybrid network, which merges two networks to provide a predictive model. In contrast to the results reported by [7], who showed that mixed deep learning is the most successful technique for forecasting stock markets when compared to other methods, this conclusion is supported by Shah et al. According to [8], while classic ARIMA models were less accurate, hybrid techniques that utilized CNN for trend identification and BiLSTM for price direction demonstrated superior performance.

These hybrid models are very proficient at catching both the quick changes that are common in intraday trading and the more general price patterns. The integration of many methods and techniques is essential for developing strong trading strategies and achieving greater profitability in the stock market.

In its entirety, although displaying promise, the process of using deep learning and hybrid models for liquidity prediction in the Middle East and North Africa's (MENA) region's liquidity prediction is still in its nascent stages and necessitates additional investigation and advancement.

III. DISCUSSION

In this paper, we analyze the hybrid CNN-biLSTM model's ability to predict liquidity in the MENA region. Our focus is on time series data from different activity sectors in various regions, including Morocco, Egypt, and Saudi Arabia.

The study experiences consecutive distinct stages, including data collection, data preprocessing, model saving, model training, model parameter tuning, and predicting results.

Figure 1 illustrates these stages in detail.

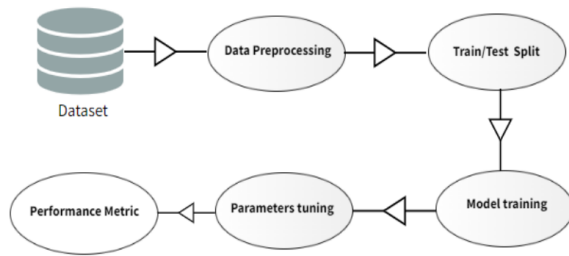


Fig. 1. Research stages

The experimental data was gathered from the official stock exchange website for each respective country, from January 1, 2020, to October 31, 2023, due to its availability on the source.

Table 1 shows the total rows of data per stock.

Country	Total rows of data
Morocco	10706
Egypt	965
Saudi Arabia	3150

TABLE I
DATASET DETAILS PER COUNTRY

The datasets include a diverse range of time-series variables. To uphold the validity and precision of the research, we implemented a filtration procedure to exclude extraneous data. This procedure enabled us to concentrate just on the pertinent variables, including the closing price, the highest price, the lowest price, the opening price, the previous day's closing price, bid-ask spread, volume, and other chronological data

sorted by date. To ensure data standardization and preserve the relative connections within different features, we implemented MinMaxScaler, which improves the convergence speed and accuracy of the model.

After that, we examine the time step parameters and set this parameter to 10, implying that the neural unit's input layer contains region data from the previous 10 days, with the volume value of the 11th day serving as the model's training target.

Then, we split the sequences and their corresponding target values into training and testing sets, where 20% of the data is reserved only for the test set, and the split is reproducible.

The model is made up of two main parts: a convolutional neural network (CNN) that handles the first inputs and a bidirectional long short-term memory (BiLSTM) that handles data in a certain order. In addition, CNN plays a crucial role in analyzing and interpreting complex patterns and features from the input datasets. Once the data is ready for processing, BiLSTM takes in the output from CNN and gains a deeper understanding of the information. Bidirectional Long Short-Term Memory (LSTM) architectures are different from traditional LSTM in that they let the model understand the sequence better by looking at both the past (backward) and the future (forward) contexts. We strategically incorporate a drop-out layer to prevent overfitting. We use Python, a widely recognized programming language, to systematically construct the intricate network structure, while Keras serves as the foundation for the neural network in this experimental study. We strategically utilize both of these languages.

Figure 2 illustrates the explicit structure of the CNN-BiLSTM model, which is based on the parameter configuration of the proposed network.

The proposed CNN-biLSTM model structure was trained on the experimental environment and parameter settings adopted in the experiment. As previously discussed, Figures 3 to 5 illustrate the prediction results, further verifying the accuracy of the model in each country.

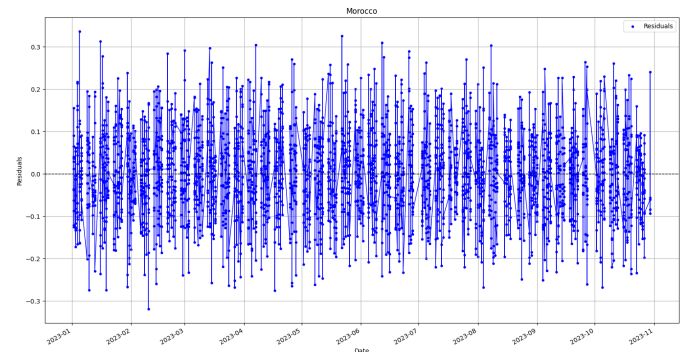


Fig. 3. Morocco: Comparison between actual and predicted value residuals

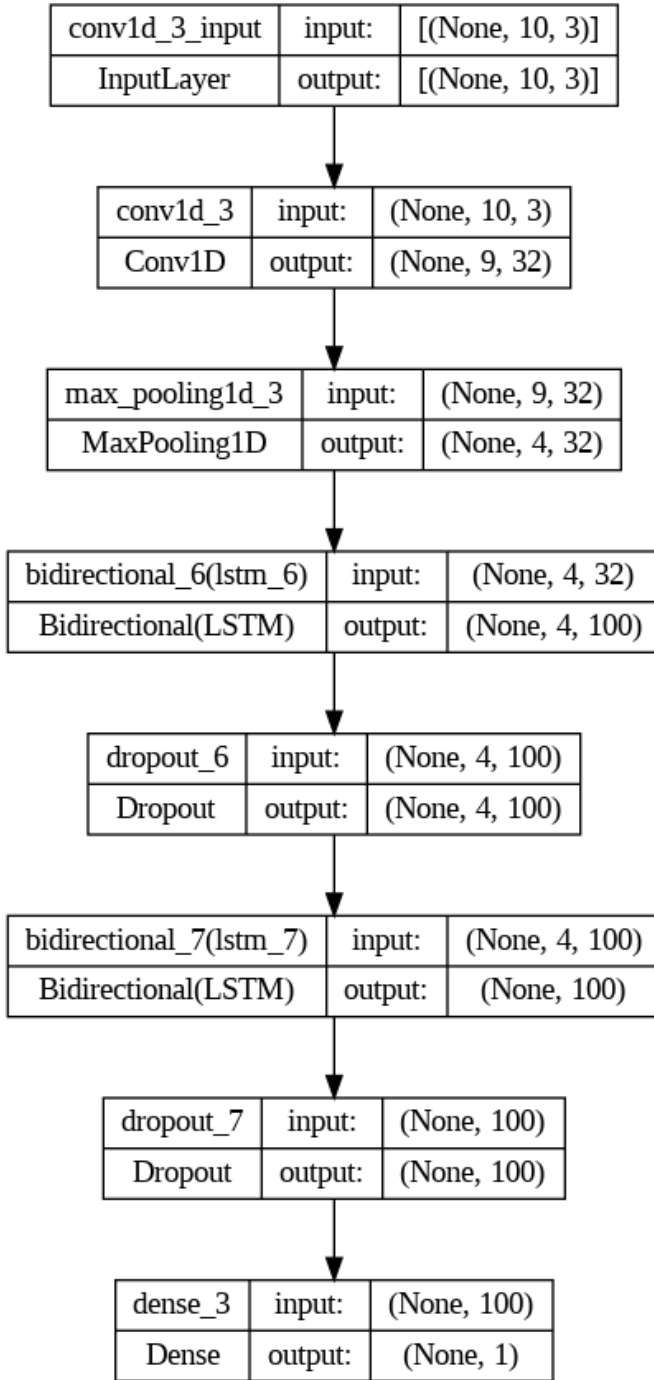


Fig. 2. CNN-biLSTM model's structure

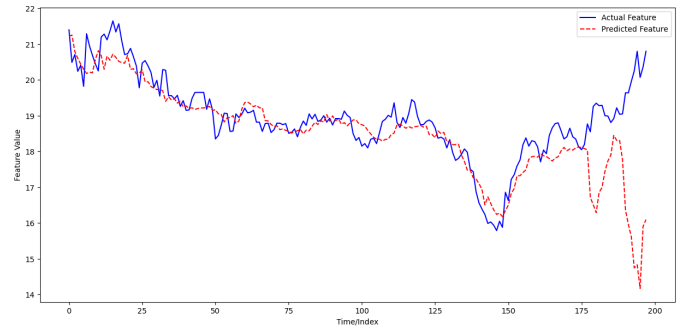


Fig. 4. Egypt: Comparison between actual and predicted value

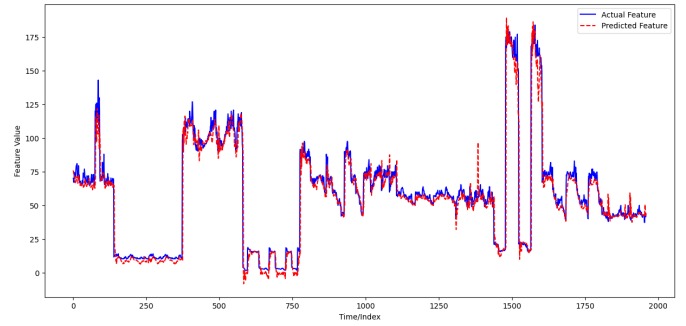


Fig. 5. Saudi Arabia: Comparison between actual and predicted value

The analysis of the disparity between actual and predicted values produced significant results on the efficacy and accuracy of our model. The figures exhibited a notable degree of concordance between predicted and actual values, suggesting a considerable level of precision in the predictions made by the model. Specifically, when considering values within the mid-range, the predictions exhibited a high degree of similarity to the actual values. Nevertheless, in the upper range, a persistent pattern of underestimation was found, with the model constantly providing lower estimations compared to the actual values. The observed trend might perhaps be ascribed to an insufficiency of adequately representative instances in our training dataset, resulting in the model's restricted familiarity with such occurrences.

The implications of these findings are significant, particularly in light of the potential use of this model for forecasting financial trends. The observed high level of accuracy within the mid-range is a promising indication of the model's potential usefulness in situations where market conditions remain relatively consistent. Nonetheless, the tendency to underestimate values at higher levels may pose significant dangers in situations of market volatility, perhaps resulting in risk evaluations that are too low.

To explicate the model's performance more succinctly, the graphical representation of the model learning plot demonstrates the progressive improvement of our process over time, owing to the acquisition of knowledge and the augmentation of expertise, as evidenced from Figure 6 to Figure 8, which

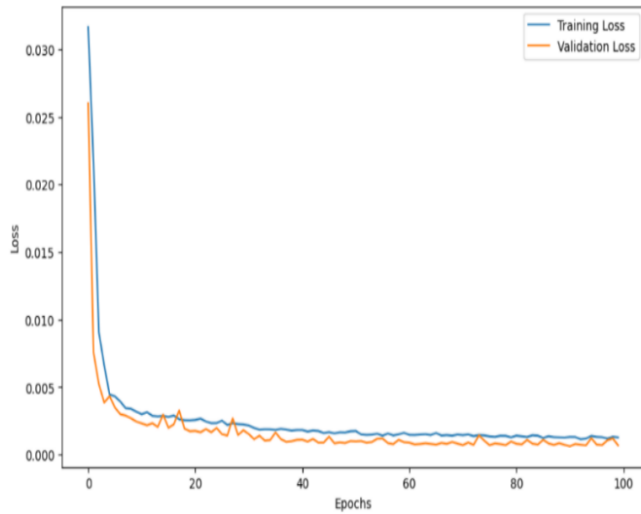


Fig. 6. Morocco; Model learning curve

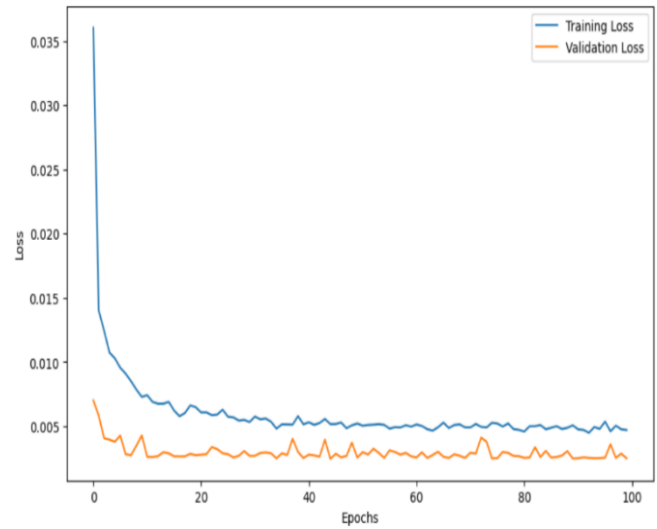


Fig. 8. Saudi Arabia: model learning curve

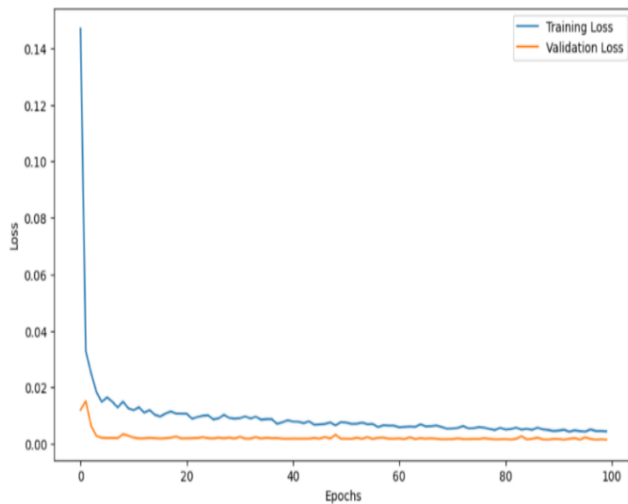


Fig. 7. Egypt: Model learning curve

provided us with a comprehensive perspective on the training dynamics and generalization capacities of the models.

The aforementioned plots provide evidence of the models' capacity to acquire knowledge from the training data and their effectiveness in handling new, unseen data, as demonstrated by the validation loss. It can be inferred from the substantial decline in both training and validation losses shown in the provided figures that the models could rapidly acquire knowledge and accurately reflect the predominant patterns found in the dataset. The first phase of the rapid acquisition of knowledge has significant importance as it establishes the foundation for the subsequent predictive capabilities of the model.

Throughout the several epochs, each model gradually ap-

proaches a state of equilibrium known as a plateau, characterized by a steady learning phase. This phenomenon arises when the model attains a state of equilibrium.

Furthermore, the observation that the models have assimilated the crucial elements of the training data serves as an indication. However, it also serves as a cautionary note, suggesting that if training continues without significant enhancements in the learning process, the models may be susceptible to overfitting. The capacity of a model to generalize is of utmost importance, especially when contemplating the use of these models in practical contexts. In this sense, Figures 7 and 10 exhibit a notable level of reliability in this aspect, as seen by their learning curves, which indicate a robust capability for consistent performance on novel data. Nevertheless, the intricate subtleties within the story lines, shown by the little disparity seen between the training and validation loss depicted in Figures 8 and 9, as well as the oscillations evident in Figure 9, indicate the need for further enhancement of the models by optimizing hyperparameters, using regularization techniques, and augmenting the dataset.

Figure 9 illustrates how our methodology was used to forecast the Moroccan market for a period of 16 years, from 2024 to 2040. At the outset, the forecast for 2024 starts with a modest figure, indicating the possibility of narrower spreads and a steady market. Nevertheless, a significant increase in 2026, followed by another surge in 2028, indicates the possibility of market disruptions or economic occurrences that might lead to higher transaction expenses and decreased liquidity. We can attribute the period of stability between 2028 and 2035, with a value of 0.10, to either market corrections

or regulatory interventions aimed at preserving financial stability. The continuing gradual rise from 2035 to 2040 may indicate projected market growth or heightened instability, necessitating proactive actions from market authorities and players.

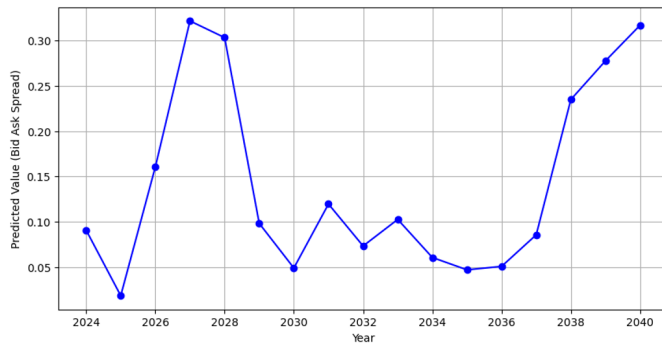


Fig. 9. Morocco : 2024-2040 prediction results

CONCLUSION

In summary, the analysis of the learning curve provides valuable perspectives on the efficacy of the MENA's region models and proposes a strategic framework for enhancing future outcomes. Through the implementation of data-driven optimizations, it is possible to gradually improve the performance of models, ensuring their effectiveness as learners and their dependability as predictors, even in the presence of uncertainties inherent in real-world data.

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